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What is a View?

SQL queries that return tables/result sets can be assigned to a **virtual table**. This is what's referred to as a view. These tables have the following characteristics.

Dynamic

Views, unlike conventional tables, **do not store data**. They **display information from other tables**. The data in a view are always fresh and up to date since the view refreshes when the tables used to generate the view change.

Data Abstraction

Views can **hide the complex data**. For example, you could join multiple tables and present only a selection of columns or rows where a column has a specific value. Views can also **restrict access to data** for different users (for example, customers, employees, suppliers, etc.).

Read-Only or Updatable

Some views are **read-only**, meaning you **can't modify the data** using the view. Others can be updatable, allowing INSERT, UPDATE, and DELETE operations on the view.



Why Do We Need Views?

Most organizations handle sensitive data such as personal, client, and employee information. Different users can be **authorized** to use or view different parts of the data and system. Here are a few examples:

Regulatory Compliance:

Ensures only **relevant** data are accessible, meeting industry **compliance** regulations.

```
CREATE VIEW CompliantView AS
SELECT CustomerID, Name, SalesDate
FROM Sales
WHERE DataCompFlag = 'Compliant';
```

Cross-Departmental Reporting:

Caters to each department's data requirements, **tailoring their reports**.

```
CREATE VIEW FinReportView AS
SELECT EmployeeID, Salary, Tax
FROM EmployeeData
WHERE Department = 'Finance';
```

Third-Party Integration:

Provides data interfaces for **external stakeholders** without revealing internal data.

```
CREATE VIEW VendorView AS
SELECT ProductID, ProductName, Stock
FROM Inventory
WHERE VendorAccess = 'Allowed';
```

CustomerID	Name	SalesDate
101	Jane Doe	2023-01-15
102	Jane Smith	2023-01-20
103	Alice Lee	2023-01-22

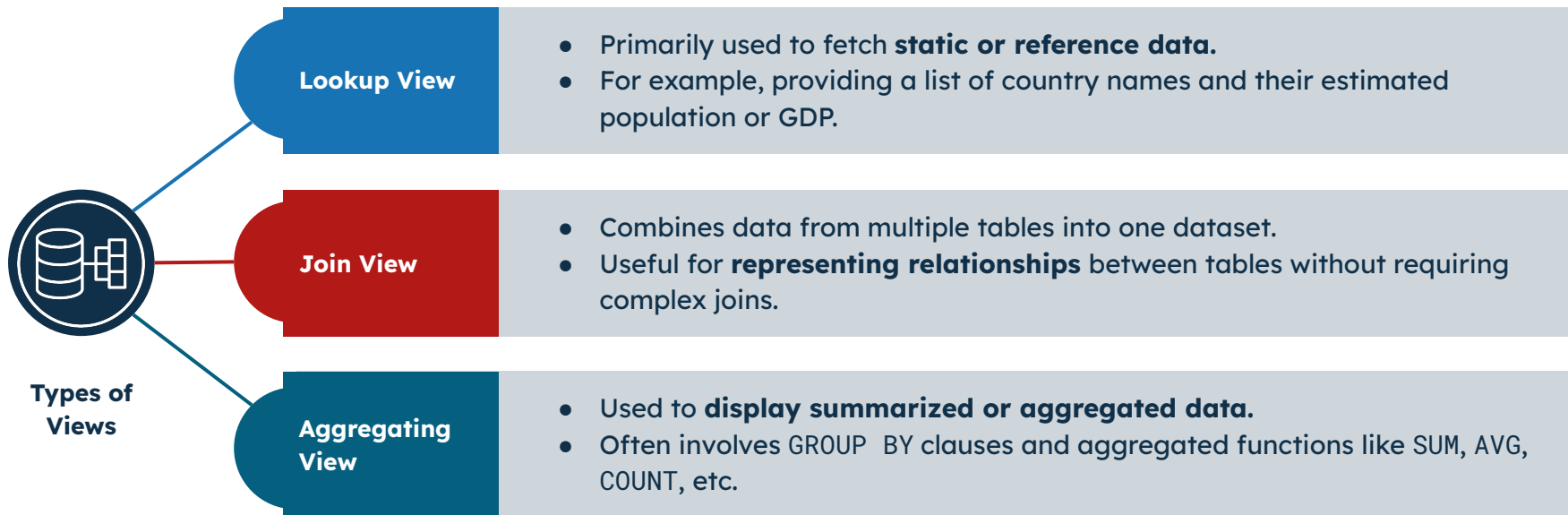
EmployeeID	Salary	Tax
E001	\$5,000	\$500
E002	\$5,500	\$480
E003	\$4,800	\$520

ProductID	ProductName	Stock
P001	Laptop	50
P002	Mobile Phone	100
P003	Mouse	200



Types of Views

While views can be broadly categorized as simple or complex, we will focus on **three specific types derived** from their **DQL query characteristics**.



Types of
Views



Lookup View Example

Let's create a lookup view that provides a **list of country names** and their **estimated population** from the "Access_to_Basic_Services" table.

```
CREATE VIEW Country_Lookup AS
SELECT
    Country_name,
    Est_population_in_millions,
    Time_period
FROM
    united_nations.Access_to_Basic_Services
```

This view, **Country_Lookup**, retrieves data from the Access_to_Basic_Services table.

It serves as a reference for **quick access** to country names and their estimated populations.

Country_name	Est_population_in_millions	Time_period
Kazakhstan	18.037	2018
Kazakhstan	18.513	2019
Kazakhstan	18.756	2020
Tajikistan	12.037	2018
Tajikistan	12.001	2019



Join View Example: Source Tables

Consider these two tables within the United Nations database:

Access_to_Basic_Services and **Regions**, which will help us to **add geographical data** to our analysis.

Table: **Access_to_Basic_Services**

Country_name	Sub_region	...	...
Sri Lanka	Central Asia	...	...
China	Eastern Asia	...	...
Canada	Northern America	...	...

Table: **Regions**

Region	Sub_region	...	...
Central and Southern Asia	Central Asia	...	...
Eastern and South-Eastern Asia	Eastern Asia	...	...
Europe and Northern America	Northern America	...	...



Join View Example

Let's create a join view that helps us to determine the **regions for each country** in our dataset. This can be done by **joining tables** Access_to_Basic_Services and Regions.

```
CREATE VIEW Region_Country_Join AS
SELECT
    A.Country_name,
    A.Sub_region,
    B.Region
FROM
    united_nations.Access_to_Basic_Services
AS A
INNER JOIN
    united_nations.Regions
AS B
ON A.Sub_region = B.Sub_region
```

This view, Region_Country_Join, retrieves data from the Access_to_Basic_Services and Regions tables. It simplifies **querying relationships** between countries and regions.

Country_name	Region	Sub_region
Sri Lanka	Central and Southern Asia	Central Asia
China	Eastern and South-Eastern Asia	Eastern Asia
Canada	Europe and Northern America	Northern America



Aggregating View Example

Let's create an aggregating view that **displays summarized data** about regions, including the total population and average GDP.

```
CREATE VIEW Region_Aggregation AS
SELECT
    Region,
    SUM(Est_population_in_millions)
        AS Total_Population_in_millions,
    AVG(Est_gdp_in_billions)
        AS Avg_GDP_in_billions
FROM
    united_nations.Access_to_Basic_Services
```

This view, Region_Aggregation, retrieves data from the Access_to_Basic_Services and Regions tables. It **provides insights** into the total population and average GDP for each region.

Region	Total_Population_in_millions	Avg_GDP_in_billions
Central and Southern Asia	11371.04644	302.7461
Eastern and South-Eastern Asia	12477.7617	1703.9799
Europe and Northern America	221.7836	550.3311



Advantages of Using Views

Data Security

Views can **restrict access** to specific columns and rows in a table, ensuring users **only see authorized data**.

Simplify Querying

Views **combine complex SQL queries**, allowing users to retrieve data without understanding or dealing with the underlying complexity.

Data Abstraction

Data is presented in a more understandable format, **abstracting the underlying table's complexity** and offering a simplified perspective of the data.

Consistency

Views **offer a consistent snapshot** of data, which can be very useful for reporting and analysis.

Reduced Network Traffic

Instead of sending multiple complex queries over a network, **applications can query a view**, potentially reducing the amount of data transferred.



Disadvantages of Using Views

Performance Overhead

Views built from multiple tables and complex conditions introduce **performance overhead** because they are run every time the view is queried.

Maintenance Complexity

As the database schema evolves, views might need updates, and if they're not managed properly, views can become **challenging to maintain**.

Potential for Misuse

Inexperienced users might treat views as physical tables, and over-reliance on views can lead to a **cluttered database with redundant views**.

Debugging

Debugging and finding the source of the issue with views, especially nested views, can be more **challenging** than with regular tables.

Updatability

Depending on the database system and the complexity of the view, we **might not be able to perform** INSERT, UPDATE, or DELETE queries directly on the view.



Best Practices When Creating Views

01. Be Specific with Columns

Instead of using `SELECT *`, **specify the exact columns** you need in the view. This might also improve performance.

02. Document Views

Always **provide comments or documentation** for views, explaining their purpose, the data they represent, and any other relevant details.

03. Avoid Nested Views

While it's possible to create a view based on another view, **excessive nesting can lead to performance issues** and increased complexity.

04. Use Schema Binding

If it's supported, use **schema binding to bind the view to the schema** of the tables used to generate it. This ensures stability and integrity.

05. Test Views Thoroughly

Before deploying a view in a production environment, **test it** to ensure it returns the expected results and performs well.

06. Stay Updated with Changes

If there are changes to the schema of the tables used to create the views, ensure that the **views are updated accordingly**.